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Hawking Radiation, Mirrors, and Horizons

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Forord

Hva skjer når et sort hull blir borte? Dette er spørsmålet som ikke bare har inspirert denne oppgava, men også min egen interesse for fysikk. Sorte hull er kanskje mest kjent for at om man først faller inn i et, kommer man aldri ut igjen. Dette er fordi Einsteins relativitetsteori setter lysets hastighet som ei øvre fartsgrense, og innafor hendelseshorisonten er tyngdekrafta så sterk at til og med lys blir dratt innover.

Hawking oppdaget i 1975 at sorte hull sender ut stråling, fordamper, og til slutt blir borte. Om ingenting slipper ut av et sort hull, hvor kommer da denne strålinga fra? Hva skjer med det som faller inn i et sort hull som deretter fordamper? I denne masteravhandlinga legger jeg fram noen av dagens modeller, og svarer ut ifra dem så godt jeg kan.

På spørsmålet om hvor strålinga oppstår, er svaret avhengig av verdien til stresstensoren utenfor et statisk objekt, for eksempel jorda. Enkelt forklart er spørsmålet hvor mye energi som befinner seg i tomrommet utenfor jordas atmosfære.

Om svaret er det som virker mest naturlig, at det er null energi i dette tomrommet, vil Hawking-stråling sendes ut direkte fra objektet under kollaps. Dette fører da til at sorte hull ikke eksisterer, fordi det ut ifra denne modellen vil være umulig å krysse hendelseshorisonten. Det er lett å forestille seg hvordan noe som faller inn mot et «sort hull» i denne modellen slipper ut igjen, siden det er utenfor hendelseshorisonten helt til det «sorte hullet» har fordampet og blitt borte.

Dersom svaret i steder er, som predikert av kvantefeltteori i krumma romtid, at energien som befinner seg i dette tomrommet er negativ, vil Hawking-stråling oppstå utenfor objektet under kollaps. Hawking-stråling oppstår da ved at like mengder positiv og negativ energi dannes, der den negative energien går innover mens den positive energien går utover. I denne modellen må ikke energien slippe ut igjen fra hendelseshorisonten, siden sorte hull fordamper ved at negativ energi faller inn. Det er dermed vanskelig å fastslå hva som skjer med noe som faller inn i et sort hull, før det fordamper og blir borte.

Et annet spørsmål, som er noe omdiskutert i litteraturen, er om hendelseshorisonten er nødvendig for at Hawking-stråling skal oppstå. I denne avhandlinga svarer jeg på dette. Uavhengig av om sorte hull fordamper ved

å sende ut positiv energi eller ved å absorbere negativ energi er svaret det samme: Det er hvor fort et objekt kollapser som avgjør hvor mye stråling som sendes ut. Det er da ikke nødvendig at en hendelseshorisont dannes for at Hawking-stråling skal oppstå. Uavhengig av kollaps vil strålinga bli nærmere inverskvadratisk proporsjonal med den totale massen til objektet, i takt med at det kollapsende objektets radius nærmer seg hendelseshorisonten, kjent som Schwarzschild-radien.

Jeg vil først og fremst takke min veileder Prof. Joakim Bergli for god hjelp med oppgava, og for å være lett å ha med å gjøre. Jeg er takknemlig for muligheten til å jobbe med et særdeles interessant emne.

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Abstract

This thesis finds that Hawking radiation does not depend on the formation of an event horizon. When an object collapses to form a black hole, it is the collapse trajectory $R(\tau)$, see section 4.1, which causes radiation to be emitted through the induced relationship between the outgoing u and ingoing v null coordinates. This is the case both by Bogolubov analysis, see section 2.2, and by computation of the stress tensor, see section 2.4. No radiation is expected from static objects like planets or static ECOs, see section 4.2. It is also found that horizon avoidance occurs when evaporation is modelled by the outgoing Vaidya metric. The horizon still forms when modelling evaporation by the ingoing Vaidya metric, see section 3.1.2. When accounting for the stress tensor outside a static object being non-zero, it is found that the ingoing Vaidya metric is appropriate for modelling black hole formation and evaporation, section 2.4.3.

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Chapter 1

Introduction

In 1975, Stephen Hawking predicted the existence of a thermal flux of energy emitted by black holes [1]. Until this time black holes were thought to be stable objects that could never shrink. If something were to fall into a black hole it would hit the singularity and presumably be destroyed. Any information about what happens to the object after it crosses the horizon is inaccessible to those who live on the outside. Combining this with the thermal evaporation of black holes, Hawking realized that this leads to a breakdown of predictability[1] in our theory [2] known as the *information loss paradox*.

I will in this thesis focus on the proposed resolution of the paradox based on horizon avoidance [3, 4, 5]. Essentially, the model posits that nothing ever becomes smaller than the Schwarzschild radius, and so we say that the event horizon never forms. If true, there is no paradox as no information is ever lost behind the horizon. This is an intuitively reasonable model. As I show in section 1.2 below, an external observer will never see something cross the event horizon. If this very observer sees the black hole evaporate in finite time, then one would be forgiven for assuming that nothing ever falls through the horizon. I discuss what condition must be met for this to be the case in chapter 3.

An important, but disputed, assumption of this model is that the event horizon itself is not necessary for the production of radiation. I show, in chapter 4, that this is indeed the case. It is how quickly something collapses which determines the production of radiation. Thermal Hawking radiation is generic to all collapse trajectories, provided they get sufficiently close to the Schwarzschild radius. This is the most important result of the thesis.

Chapter 2 is a summary of important results and techniques relevant to understanding black hole evaporation. Most importantly, the relationship between the null coordinates u and v, and also the calculation of the renormalized stress tensor in curved spacetime. In chapter 5, I mention some

[1] No S-matrix describes the process of black hole formation and evaporation.

contemporary areas of research I find especially fascinating.

The rest of the introduction is dedicated to simple derivations of some relevant properties of black holes. I find the mass of an evaporating black hole as a function of time, and I show how time dilation makes crossing the horizon take an infinite amount of time to an external observer.

1.1 Basics of black hole evaporation

As we will soon see, section 2.2, a black hole of mass M will emit radiation with the Hawking temperature

$$T_{\mathrm{H}} = \frac{\hbar c^3}{G k_{\mathrm{B}}} \frac{1}{8\pi M}. \tag{1.1}$$

$\hbar$ is the reduced Planck's constant, c is the speed of light, G is Newton's gravitational constant, and k_{B} is Boltzmann's constant. As these constants tend to get in the way when doing calculations, I will adopt what is known as natural units for the majority of the thesis. This means that all such constants will be set equal to one. In this case the temperature is given by $T_{\mathrm{H}} = 1/8\pi M$, which could be interpreted as measuring temperature in units of kg^{-1}. This should not be taken too seriously and simply tells us that we must add back the constants in such a way that the units make sense. For a solar-mass black hole ($M_{\odot} = 2 \times 10^{30}\mathrm{kg}$), The temperature is about 60nK.

The radius of a black hole is given by its Schwarzschild radius $r_s = 2GM/c^2$ [6], whence its area is

$$A = 4\pi r_s^2 = \frac{G^2}{c^4} 16\pi M^2.$$

By the Stefan-Boltzmann law, the power radiated by a black hole is

$$P = A\sigma T_{\mathrm{H}}^4 = \frac{\hbar c^6}{G^2} \frac{1}{15360\pi M^2}. \tag{1.2}$$

Where the following expression for the Stefan-Boltzmann constant has been used:

$$\sigma = \frac{k_{\mathrm{B}}^4}{c^2 \hbar^3} \frac{\pi^2}{60}.$$

The evaporation of the black hole can then be modelled, making use of Einstein's famous equation $E = Mc^2$, as

$$\frac{dM}{dt} = -\frac{\hbar c^4}{G^2} \frac{1}{15360\pi M^2}.$$

The solution is

$$M(t) = \sqrt[3]{M_0^3 - \frac{\hbar c^4}{G^2} \frac{t}{5120\pi}},$$

where $M_0 = M(0)$ is the initial mass of the black hole.

The time it takes for a black hole to evaporate completely is then proportional to the cube of its mass

$$t_{\text{ev}} = \frac{G^2}{\hbar c^4} 5120\pi M_0^3.$$

A black hole with the mass of our sun will then take 2×10^{67} years to evaporate. For comparison, the age of the universe is thought to be 13.8×10^9 years. The observation of the evaporation of a (stellar sized) black hole would then, somewhat ironically, be evidence of this model being deeply flawed.

1.2 Falling through the event horizon

An interesting quality of Einstein's theory of general relativity is the phenomenon of gravitational time dilation. A clock close to a gravitating object will appear to run slowly as seen by an observer far away. This time dilation goes to infinity at the event horizon of a black hole. A consequence of this is that crossing the event horizon will take a finite amount of time for someone falling into a black hole. However, an external observer will say that crossing the horizon takes an infinite amount of time. See Carroll [6] chapter 5 for a textbook discussion.

Let us see why this is. Begin with the radial part of the Schwarzschild line element, given by

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{r_s}{r}}. \tag{1.3}$$

r_s is the Schwarzschild radius which is the position of the event horizon.

Take an inn-falling observer with the trajectory $R = -v\tau$, with τ his proper time. The position of the inn-falling observer as described by an observer far away is given by

$$x^\mu = \begin{pmatrix} t \\ R(\tau(t)) \end{pmatrix},$$

and thus the four-velocity is given by

$$\dot{x}^\mu = \begin{pmatrix} \dot{t} \\ \dot{R}(\tau(t)) \end{pmatrix}.$$

Dot here signifies the derivative with respect to the proper time τ.

Normalization[2] of the four-velocity yields the equation

$$C(r)\dot{t}^2 - C(r)^{-1}\dot{R}^2 = 1, \tag{1.4}$$

[2] $\dot{x}^2 = -1$.

where C is the Schwarzschild factor

$$C(r) = 1 - \frac{r_s}{r}. \tag{1.5}$$

From (1.4) we get the equation for the time dilation

$$dt = \frac{\sqrt{C(R) + \dot{R}^2}}{C(R)} d\tau.$$

This blows up for $R = r_s$. Even though τ remains finite across the horizon, t goes to infinity. The interpretation is that gravitational time dilation makes the finite proper time it takes to cross the horizon infinitely long for an external observer.

Chapter 2

Theory

Here I introduce some of the concepts that are useful for understanding the content of this thesis. First I introduce Penrose diagrams, also known as conformal diagrams, which are useful for visualising the path of light through, and the causal structure of, various spacetimes. Next, I give a brief derivation of the Hawking effect using Hawking's original approach leveraging Bogolubov transformations. Then I go on to discuss the toy model of a moving mirror, also leveraging Bogolubov transformations. I conclude the chapter by calculating the vacuum expectation value of the renormalized stress tensor of a massless scalar field, first in the case of a moving mirror and then in curved spacetime.

2.1 Penrose diagrams

A Penrose diagram is a conformal diagram that allows us to see all of (a two-dimensional) spacetime in a single diagram while also keeping the causal structure obvious. For a textbook discussion see Carroll [6] appendix H. Taking two-dimensional Minkowski space, the line element is

$$ds^2 = -dt^2 + dx^2.$$

Because we care about the causal structure, we make the switch to null[1] coordinates u and v defined by

$$du = dt - dx,$$
$$dv = dt + dx.$$

For $du = 0$ (constant u) we have a right-moving light ray, and for $dv = 0$ we have a left-moving light ray. In this coordinate system, the line element is

$$ds^2 = -dudv. \tag{2.1}$$

[1] Null refers to things that move at the speed of light; they follow paths of zero proper time.

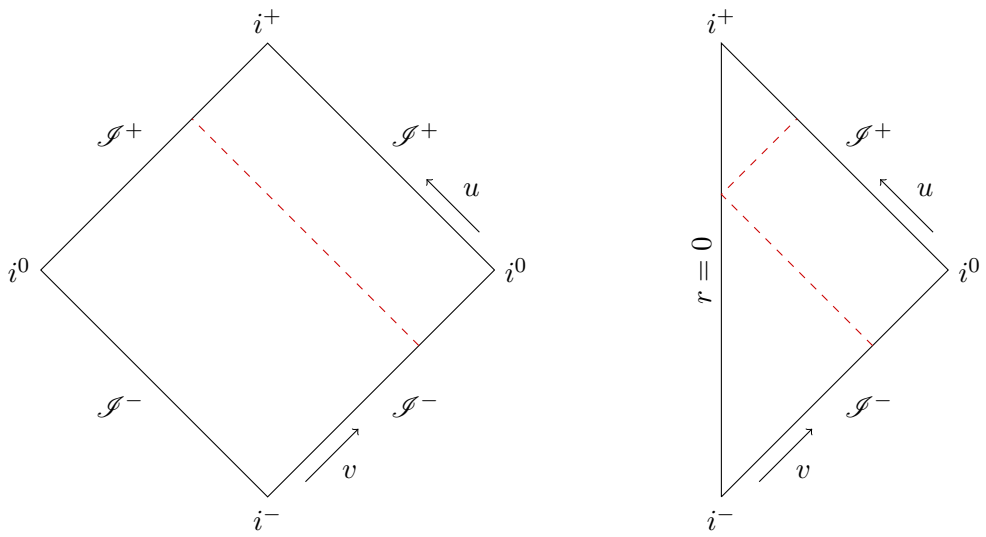

Figure 2.1: Penrose diagrams of two-dimensional Minkowski space in cartesian coordinates (left) and radial coordinates (right). $i^{\pm}$ are time-like infinities, while i^0 is space-like infinity. Most interesting for this thesis are the null-infinities $\mathscr{I}^{\pm}$ where light rays begin and end. The dashed line shows the path of a light ray.

Now define coordinates $\bar{u}, \bar{v} \in \langle -\pi/2, \pi/2 \rangle$:

$$u = \tan \bar{u},$$
$$v = \tan \bar{v}.$$

The line element is now

$$ds^2 = -\left(\sec^2 \bar{u} \sec^2 \bar{v}\right) d\bar{u} d\bar{v}.$$

We now have finite coordinates representing all of spacetime, and light moves along lines of constant $\bar{u}$ or constant $\bar{v}$.

In the diagram on the right in figure 2.1 we see the resulting Penrose diagram in radial coordinates. The use of radial coordinates changes the meaning of right-moving and left-moving to outgoing and incoming respectively. v then parameterizes past null-infinity $\mathscr{I}^-$ from $-\infty$ at past time-like infinity i^- to ∞ at space-like infinity i^0, while u parametrizes $\mathscr{I}^+$ from $-\infty$ at i^0 to ∞ at i^+.

Looking at the figure, it is possible to make a connection between u and v using the light rays. An incoming light ray (with constant v) will be "reflected" off $r = 0$ and continue as an outgoing light ray (with constant u). In this case, the relationship is given by

$$u = v + \text{const.}. \tag{2.2}$$

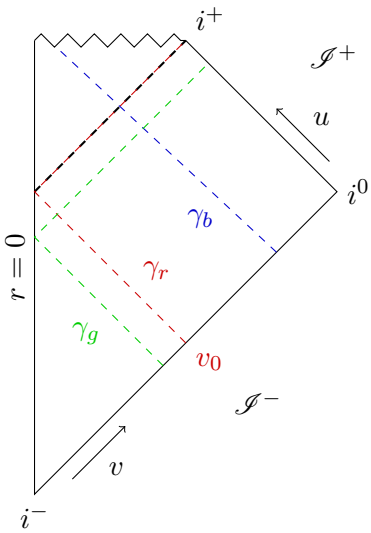

Figure 2.2: A Penrose diagram of matter collapsing to a black hole. The red light ray γ_r shows the path of the last light ray to escape to $\mathscr{I}^+$. While incoming it is labelled by $v = v_0$. All light rays like γ_g with $v < v_0$ escape to $\mathscr{I}^+$, and all light rays like γ_b with $v > v_0$ end on the singularity and never reach $\mathscr{I}^+$.

The way this connection changes in the presence of a black hole is what allowed Hawking to derive his famous result in 1975 [1].

In figure 2.2 we see a Penrose diagram of matter collapsing to form a black hole. The connection between u and v is clearly different. Only $v < v_0$ are related to u. We say that modes with $v > v_0$ are lost behind the horizon, and the map $u(v) : v \mapsto u$ is not bijective. If this non-bijectivity is essential for Hawking radiation, then a horizon is also essential. If there is Hawking radiation even in geometries where $u(v)$ is bijective, then the horizon is not essential.

2.2 Hawking radiation

Hawking radiation is the process, first derived by Hawking in 1975 [1], by which black holes are thought to lose mass and eventually evaporate completely. I will give here a brief overview of the derivation, for a textbook treatment see either Parker and Toms [7] chapter 4 or Birrell and Davies [8] chapter 8.

Consider a spacetime with a spherically symmetric distribution of dust with mass M which collapses to form a black hole. In the asymptotic past assume that the dust is spread sparsely throughout the universe such that

we effectively have Minkowski space. In null coordinates the metric[2] is (2.1)

$$ds^2 = -dudv.$$

We expect Hawking radiation to be dominant in the radial direction. Given the spherical symmetry of the Schwarzschild spacetime, it is reasonable to assume that the radiation is also spherically symmetric. This is most easily achieved through radially emitted radiation. The final results will not differ qualitatively from the full four-dimensional case, see Birrell and Davies [8] section 8.1 for a more rigorous argument.

Hawking radiation being a quantum phenomenon, we let there live in this universe a quantized, massless, real, minimally coupled scalar field ϕ. We pick this one out of all the possible quantum field theories because it is the simplest. Its equation of motion is the Klein-Gordon equation [8]

$$\Box\phi = \partial_u\partial_v\phi = 0.$$

The solution to which can be expanded in positive and negative frequency modes as

$$\phi = \int_0^\infty d\omega \left(f_\omega a_\omega + f_\omega^* a_\omega^\dagger \right). \tag{2.3}$$

a_ω and $a_\omega^\dagger$ are respectively annihilation and creation operators for quanta of frequency ω, and satisfy the canonical commutation relations $[a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega')$. The modes f_ω take the form

$$f_\omega \sim e^{-i\omega u} + e^{-i\omega v}.$$

Normalizing the Klein-Gordon inner product on a constant t hypersurface [8]

$$(f_\omega, f_{\omega'}) = -i \int_0^\infty dr \left(f_\omega^* \partial_t f_{\omega'} - f_\omega' \partial_t f_\omega^* \right) = \delta(\omega - \omega'),$$

yields

$$f_\omega = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega u} + \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega v}. \tag{2.4}$$

In the asymptotic future, when the black hole has formed, we will have the Schwarzschild metric (1.3)

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{r_s}{r}}.$$

Where $r_s = 2M$ is the Schwarzschild radius of a black hole of mass M.

[2] The metric is the matrix $g_{\mu\nu}$ such that the line element is given by $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.

Null coordinates now take the form

$$du = dt - \frac{dr}{1 - \frac{r_s}{r}}, \tag{2.5a}$$

$$dv = dt + \frac{dr}{1 - \frac{r_s}{r}}, \tag{2.5b}$$

in which coordinates the metric is given by

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dudv, \tag{2.6}$$

The d'Alembertian operator in curved spacetime takes the form [8]

$$\Box\phi = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi),$$

where g is the metric determinant and $g^{\mu\nu}$ is the inverse metric.

In null coordinates in Schwarzschild spacetime, the Klein-Gordon equation is given by

$$\Box\phi = \frac{4}{1 - \frac{r_s}{r}}\partial_u\partial_v\phi = 0.$$

This has the same solutions as in flat space $\partial_u\partial_v\phi = 0$, and we can therefore expand the field ϕ in different modes p_ω given by

$$p_\omega = \frac{1}{\sqrt{4\pi\omega}}e^{-i\omega u} + \frac{1}{\sqrt{4\pi\omega}}e^{-i\omega v}. \tag{2.7}$$

A Bogolubov transformation relates the past and future sets of modes,

$$p_\omega = \int_0^\infty d\omega' \left(\alpha_{\omega\omega'} f_{\omega'} + \beta_{\omega\omega'} f^*_{\omega'}\right). \tag{2.8}$$

The Bogolubov coefficient $\beta_{\omega\omega'}$ determines how much positive and negative frequencies mix between different bases and directly determines particle production [8].

An incoming wave $e^{-i\omega v}$ from past null infinity will pass through $r = 0$ and become an outgoing wave $e^{-i\omega u}$ ending on future null infinity provided it is early enough to escape the singularity ($v < v_0$). Because of this relationship between $e^{-i\omega v}$ in the past and $e^{-i\omega u}$ in the future, the relationship between f_ω and p_ω is not as trivial as one is led to believe by (2.4) and (2.7). I determine the function relating v and u in (4.4) in the case of a collapsing null shell, see Birrell and Davies [8] for the more general case of a collapsing ball. We have

$$u(v) = v - 2r_s \ln(v_0 - v) + \text{const.}. \tag{2.9}$$

Close to the horizon ($v \approx v_0$), the logarithm dominates and it is common to ignore the linear term in calculations, see Hawking's original paper [1]. This

is because we are interested in the far future $u \to \infty$, which in figure 2.2 is close to the horizon. We also take $e^{-i\omega u(v)}$ to be 0 for $v > v_0$ as these modes hit the singularity and do not reach $\mathscr{I}^+$. Using (2.8) to relate the incoming $e^{-i\omega v}$ modes to the outgoing $e^{-i\omega u(v)}$ modes and Fourier transforming yields [8]

$$\beta_{\omega\omega'} = \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\int_{-\infty}^{v_0} dv e^{-i\omega' v} e^{-i\omega u(v)}. \tag{2.10}$$

This integral, with $u(v)$ given by (2.9), can be evaluated exactly using contour integration. The result is (ignoring phase factors) [8]

$$\beta_{\omega\omega'} = \frac{r_s}{\pi}\sqrt{\frac{\omega}{\omega'}} e^{-\pi r_s \omega}\Gamma(i2r_s\omega).$$

Thus,

$$|\beta_{\omega\omega'}|^2 = \frac{r_s}{\pi\omega'}\frac{1}{e^{\beta\omega}-1}, \tag{2.11}$$

with $\beta = 4\pi r_s$. For convenience, I give the known identity:

$$|\Gamma(ix)|^2 = \frac{2\pi}{x}\frac{1}{e^x - e^{-x}}.$$

We recognise the Bose-Einstein distribution in (2.11) and notice that the black hole has temperature $T = \beta^{-1} = 1/4\pi r_s = 1/8\pi M$. This is done formally by considering wave packets, see Parker and Toms [7] chapter 4 for a discussion.

2.3 Moving mirrors

A simple model, due to Fulling and Davies [9], which captures the physics of Hawking radiation quite neatly is the moving mirror model. See Birrell and Davies [8] for a textbook discussion. In two-dimensional Minkowski space, a mirror is modelled as a reflecting point with the boundary condition that all fields vanish. Let the mirror move according to the function $z(t)$, i.e.

$$x = z(t).$$

Light rays reflecting off the mirror give a relationship between the null coordinates u and v depending on the trajectory of the mirror. Typically it is easier to start with a function that relates u and v and solve for the trajectory which generates it.

2.3.1 Hawking trajectory

Let us consider a mirror moving on a trajectory such that it generates Hawking radiation. The function $u(v)$ from the black hole calculation is (4.4)

$$u(v) = v - 2a\ln(v_0 - v). \tag{2.12}$$

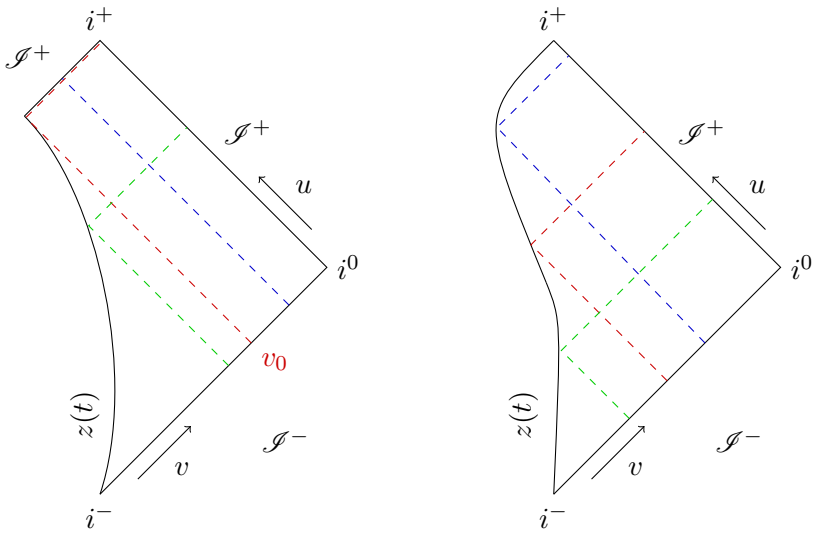

Figure 2.3: Penrose diagrams of moving mirrors. On the left, we see the mirror from (2.13). There is a horizon in the sense that light rays later than v_0 are never reflected and instead end up on left $\mathscr{I}^+$. On the right, we see the mirror from (2.15) which does not accelerate fast enough to end up on left $\mathscr{I}^+$. There is no horizon, and all light rays are reflected by the mirror.

Taking the null coordinates to be $u = t - x$ and $v = t + x$, we get

$$v - u = 2x,$$

from which the trajectory that generates Hawking radiation is given by

$$z(t) = a \ln (v_0 - t - z(t)), \tag{2.13}$$

where a determines how sharply the mirror becomes light-like. The trajectory is that of a mirror coming slowly from the right before accelerating and hitting left $\mathscr{I}^+$. The diagram on the left of figure 2.3 shows this trajectory. The presence of a horizon makes the non-bijectivity of $u(v)$ easy to see. Since $u(v)$ is the same as for Hawking radiation ($\beta_{\omega\omega'}$ is also the same in this case) we get a thermal spectrum of particles created. The Bogolubov analysis is identical for moving mirrors and collapsing objects [8].

2.3.2 Horizonless trajectory

The question of whether a horizon is necessary for Hawking radiation can now be made explicit by considering a mirror moving without generating a horizon. In the Penrose diagram, this mirror would end up at i^+ instead of left $\mathscr{I}^+$. I have not seen anyone calculate the Bogolubov coefficients for

such a trajectory. This is presumably because the integral (2.10) is difficult to solve for arbitrary $u(v)$. I will therefore do this calculation in a specific case. Consider the relationship between u and v given by

$$u(v) = v + 2e^{av}, \tag{2.14}$$

generated by the trajectory

$$z(t) = -e^{a(t+z(t))}. \tag{2.15}$$

a is also here related to the acceleration of the mirror. A conformal diagram depicting a mirror with this trajectory is seen on the right of figure 2.3. Here there is no horizon, and $u(v)$ is indeed a bijective function. Hawking radiation is again determined by $\beta_{\omega\omega'}$, which is given by (cf. (2.10))

$$\beta_{\omega\omega'} = \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\int_{-\infty}^{\infty} dv e^{-i\omega' v} e^{-i\omega u(v)}.$$

The integral is not cut short, as there is no horizon and $u(v)$ is defined for all v. It can be evaluated exactly using contour integration, and the result is

$$\beta_{\omega\omega'} = \frac{e^{-\pi(\omega'+\omega)/2a}}{2\pi a}\sqrt{\frac{\omega'}{\omega}}\Gamma(-i(\omega'+\omega)/a).$$

Thus

$$|\beta_{\omega\omega'}|^2 = \frac{\omega'}{2\pi a\omega(\omega'+\omega)}\frac{1}{e^{2\pi(\omega'+\omega)/a}-1}.$$

The result $\beta_{\omega\omega'} \neq 0$ demonstrates the existence of Hawking-like radiation even in the absence of a horizon. It does however differ from (2.11) in the sense that it is not thermal. Most importantly it lends credence to evaporation still taking place without the formation of an event horizon.

The energy emitted as a function of u due to these mirror trajectories is shown in figure 2.4 in terms of the stress tensor component T_{uu}. I show how this is calculated in section 2.4.

2.3.3 The function $u(v)$

It is hopefully clear that the production of Hawking radiation is governed by the function $u(v)$.

If a mirror moves at constant velocity, we expect no particle production due to relativity. A mirror moving with constant velocity gives a linear

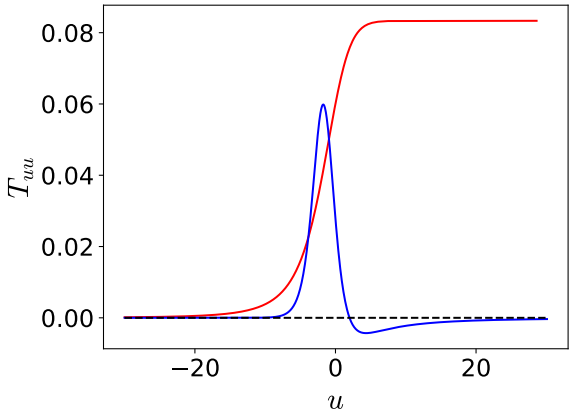

Figure 2.4: Here we see the stress tensor component T_{uu} due to the motion of moving mirrors. The results have been scaled up by a factor of 16π (cf. (2.19)). The red is due to (2.12) representing Hawking radiation with $v_0 = 0$. The blue is due to (2.14) representing a mirror trajectory with no horizon. In both cases the parameter $a = 1$. It is clear that both trajectories produce radiation, however, only the one with a horizon radiates indefinitely. We also see evidence of the emission of negative energy, however, it is clear from the figure that the total amount of radiated energy is positive.

relationship for $u(v)$. Taking $u(v) = av + b$ gives

$$\begin{aligned}
\beta_{\omega\omega'} &= \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\int_{-\infty}^{\infty} dv e^{-i\omega' v} e^{-i\omega u(v)} \\
&= \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\int_{-\infty}^{\infty} dv e^{-i\omega' v} e^{-i\omega(av+b)} \\
&\propto \delta(a\omega + \omega'),
\end{aligned}$$

with $a > 0$. $\delta(x)$ is the Dirac delta function, and because the frequency is positive will $\beta_{\omega\omega'} = 0$. There is no particle production in this case.

In the case of a static star or planet $u(v) = v + \text{const.}$, and thus this analysis predicts no Hawking radiation from these objects. See section 4.2.

When dust collapses to a static star or planet, $u(v)$ will only be non-linear during collapse. We therefore expect Hawking radiation during collapse, but not asymptotically as for a black hole.

The reason we see Hawking radiation asymptotically for a black hole is that the horizon generates a non-linear $u(v)$ even for large u. This is because collapse is an eternal process for an external observer, see section 1.2.

2.4 The stress tensor

Calculating the Bogolubov coefficient $\beta_{\omega\omega'}$ only gives information about the total amount of radiation. The stress tensor $T_{\mu\nu}$, being a local quantity, gives information about where, when, and how much radiation is produced. I will, in this section, give a derivation of the stress tensor of a scalar field in the presence of a moving mirror as well as for a collapsing object.

It is well known that the vacuum expectation value of the stress tensor $\langle T_{\mu\nu}\rangle$[3] of a scalar field is divergent even in flat spacetime, see Schwartz [10] section 12.5 for an introductory textbook discussion. In the case of a massless, real, minimally coupled scalar field ϕ, it is given by

$$\langle 0| T_{\mu\nu} |0\rangle = \langle 0| \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}\partial_\alpha\phi\partial^\alpha\phi |0\rangle \,, \tag{2.16}$$

see Birrell and Davies chapter 6. In null coordinates u and v, this is diagonal. The off-diagonal terms vanish due to the metric tensor $g_{\mu\nu}$ being off-diagonal (cf. (2.1)). This is a consequence of the stress tensor being traceless for a conformal field theory, which is what we are studying. See Qualls [11] for a discussion. In null coordinates, the trace is proportional to the off-diagonal components as we have $g^{\mu\nu}T_{\mu\nu} = 2g^{uv}T_{uv}$. The two diagonal terms are still divergent.

The theory predicting an infinite energy density everywhere in empty space is nonsensical, and the theory must be modified to fit the real world. This is done by renormalization. This entails subtracting off the infinite contribution in order to be left with a finite, hopefully reasonable, result.

For the vacuum state (in flat and empty space) we simply postulate that $\langle T_{\mu\nu}\rangle_{\text{ren}} = 0$. The subscript indicates that it has been renormalized.

In the case of a mirror moving through empty space, the mirror itself will emit radiation as seen in section 2.3. To calculate the stress tensor we should not simply postulate that it vanishes, but rather subtract off the divergent stress tensor due to a stationary mirror trajectory. This is because we expect the stress tensor to vanish in the case of a stationary mirror. We will then be left with a finite result $\langle T_{\mu\nu}\rangle_{\text{ren}}$ that only describes the energy emitted by the mirror. If the surrounding space is not a vacuum, i.e. $T_{\mu\nu} \neq 0$, the mirror will still emit the same radiation. The resulting stress tensor will then be the sum of the background $T_{\mu\nu}$ and the renormalized contribution due to the motion of the mirror. This was originally done by Fulling and Davies [9] in 1976.

The situation is identical for a collapsing object. $\langle T_{\mu\nu}\rangle_{\text{ren}}$ will be determined by the sum of the background stress tensor, known as "vacuum

[3] I may refer to this as just the stress tensor, the meaning should be clear from the context.

polarization", and a contribution due to collapse. This was achieved the same year by the same authors in collaboration with Unruh [12].

2.4.1 The stress tensor in the presence of a moving mirror

The renormalized expectation value of the stress tensor in the presence of a moving mirror can be calculated by point-splitting regularization and renormalized by subtraction of the stress tensor due to a stationary mirror. See Fulling and Davies [9]. I will closely follow the approach by Kawai *et al.* [3].

The field ϕ can be decomposed into field modes as (cf. (2.3))

$$\phi = \int_0^\infty d\omega \left(f_\omega a_\omega + f_\omega^* a_\omega^\dagger \right), \tag{2.17}$$

where now

$$f_\omega = \frac{1}{\sqrt{4\pi\omega}} \left(e^{-i\omega v} - e^{-i\omega u} \right), \qquad \text{stationary mirror;} \tag{2.18a}$$

$$f_\omega = \frac{1}{\sqrt{4\pi\omega}} \left(e^{-i\omega v} - e^{-i\omega f(u)} \right), \qquad \text{moving mirror.} \tag{2.18b}$$

The first equation satisfies $\phi(z(t)) = 0$ on the mirror $z(t) = 0$, because $v - u = 2z(t)$ here implies $e^{-i\omega v} - e^{-i\omega u} = 0$ at $x = z(t)$. In the second equation we have $z(t)$ an arbitrary trajectory of the mirror, and $v - u = 2z(t)$ implies $e^{-i\omega v} - e^{-i\omega f(v-2z(t))} = 0$ at $x = z(t)$. At $x = z(t)$ we have $v = f(u)$ which is the inverse function of the previously encountered $u(v)$. When there is no chance of confusion, I will refer to $f(u)$ simply as $v(u)$.

Calculating $\langle T_{\mu\nu} \rangle$ for a moving mirror is now straightforward. The result is, as previously mentioned, infinite. The expression can, however, be renormalized by subtracting the divergence also present for a stationary mirror. As previously mentioned, the off-diagonal components vanish even without renormalization, and now $\langle T_{vv} \rangle_{\text{ren}} = 0$ as it does not depend on the motion of the mirror. That is, $\langle T_{vv} \rangle$ is identical regardless of the mirror's trajectory from (2.18) and (2.16).

The only non-zero component to be determined is $\langle T_{uu} \rangle_{\text{ren}}$. This can be done by point-splitting regularization as[4]

$$\langle 0 | T_{uu} | 0 \rangle = \lim_{u' \to u} \langle 0 | \partial_u \phi(u) \partial_{u'} \phi(u') | 0 \rangle .$$

[4] $g_{uu} = 0$.

We get, taking $u' = u + i\varepsilon$:

$$
\begin{aligned}
&\langle 0| \partial_u \phi(u) \partial_{u'} \phi(u') |0\rangle \\
&= \frac{1}{4\pi} \int d\omega \, d\omega' \sqrt{\omega\omega'} \dot{f}(u+i\varepsilon) \dot{f}(u) e^{-i\omega f(u)} e^{i\omega' f(u+i\varepsilon)} \langle 0| a_\omega a^\dagger_{\omega'} |0\rangle \\
&= \frac{1}{4\pi} \int d\omega \, \omega \left(\dot{f} + i\varepsilon \ddot{f} - \varepsilon^2 \frac{\dddot{f}}{2} + \mathcal{O}(\varepsilon^3) \right) \dot{f} e^{-\omega \dot{f} \varepsilon} \left(1 - i\varepsilon^2 \omega \frac{\ddot{f}}{2} + \varepsilon^3 \omega \frac{\dddot{f}}{6} - \varepsilon^4 \omega^2 \frac{\ddot{f}^2}{8} + \mathcal{O}(\varepsilon^4) \right) \\
&= \frac{1}{4\pi} \left(\frac{1}{\varepsilon^2} + \frac{1}{4} \left[\frac{\ddot{f}^2}{\dot{f}^2} - \frac{2}{3} \frac{\dddot{f}}{\dot{f}} \right] + \mathcal{O}(\varepsilon) \right),
\end{aligned}
$$

where $f = f(u)$. Renormalizing by subtracting the stress tensor for a stationary mirror

$$
\langle 0| T_{uu}(u) |0\rangle_{\text{ren}} = \lim_{u' \to u} \left[\langle 0| \partial_u \phi(u) \partial_{u'} \phi(u') |0\rangle - \langle 0| \partial_u \phi_0(u) \partial_{u'} \phi_0(u') |0\rangle \right],
$$

yields

$$
\langle T_{uu} \rangle_{\text{ren}} = \frac{1}{16\pi} \mathcal{F}(\dot{f}; u), \tag{2.19}
$$

with the useful definition

$$
\mathcal{F}(f; u) := \frac{\dot{f}^2}{f^2} - \frac{2}{3} \frac{\ddot{f}}{f}.
$$

It is then clear that there is no particle production for linear $f(u)$ ($u \propto v$), in agreement with the Bogolubov analysis, see section 2.3. We also see that there is particle production only while $f(u)$ is non-linear, as expected from the Bogolubov analysis. Most worryingly there seems to be $f(u)$ for which $\langle T_{uu} \rangle_{\text{ren}} < 0$, i.e. the emission of negative energy. In total, however, the mirror always produces more positive than negative energy for all physically reasonable trajectories. It is also the case that the condition for non-zero $T_{\mu\nu}$ is that $\dddot{z} \neq 0$, constant acceleration is not sufficient [9]. For the stress tensor due to specific trajectories see Akal *et al.* [13].

2.4.2 The stress tensor due to collapsing matter

Nowhere in the derivation of the stress tensor given above, was the assumption that we were dealing with a moving mirror essential. The essential features were the decomposition into field modes f_ω (2.18) and the relationship between null coordinates u and v. As seen in section 2.2, a collapsing object will also induce a relation between the null coordinates, and the field modes take the same basic form (2.7). This means that the calculation of the stress tensor due to collapse will proceed the same way as for a moving mirror. This is what Kawai *et al.* [3] do.

See (4.4) for the relationship between u and v for matter collapsing to a black hole. Considering the late stages of collapse ($v \approx v_0$) we are safe to ignore the linear term, and we readily obtain

$$f(u) = v(u) \sim e^{-\frac{u}{2r_s}},$$

where I use $\sim$ to indicate the part of the expression that will contribute to the stress tensor. From (2.19), this yields

$$\langle T_{uu} \rangle_{\text{ren}} = \frac{1}{192\pi r_s^2} = \frac{1}{768\pi M^2}. \tag{2.20}$$

Assuming that the stress tensor is zero outside of a static body like a planet, this will be the full stress tensor during collapse[5]. According to this model, a collapsing shell will emit null dust, and the outgoing Vaidya metric will describe evaporation [3, 4, 5]. This will be discussed in chapter 3. The stress tensor describes outgoing null dust which must originate at the shell, or at the horizon in the case of an evaporating black hole.

In the next section, we will see that the stress tensor outside a planet is non-zero, which will alter this picture and have significant implications for collapse.

2.4.3 Vacuum polarization and the complete stress tensor

Taking the approach of Massar *et al.* [14], I will now argue that the stress tensor should not vanish outside a static gravitating object. Let us begin with a general metric in null coordinates

$$ds^2 = -C(u,v)dudv.$$

By a coordinate transformation, we obtain in different null-coordinates $\tilde{u}(u)$ and $\tilde{v}(v)$

$$ds^2 = -C(u,v)\frac{du}{d\tilde{u}}\frac{dv}{d\tilde{v}}d\tilde{u}d\tilde{v},$$

where $u(\tilde{u})$ and $v(\tilde{v})$ are the inverse functions. A massless scalar field has modes given by[6]

$$f_\omega = \frac{1}{\sqrt{4\pi\omega}}\left(e^{-i\omega v} - e^{-i\omega u}\right), \qquad (u,v);$$
$$f_\omega = \frac{1}{\sqrt{4\pi\omega}}\left(e^{-i\omega \tilde{v}} - e^{-i\omega \tilde{u}}\right), \qquad (\tilde{u},\tilde{v}).$$

This is just a more general version of (2.18). For a moving mirror only the null coordinate u changed $u \to f(u)$ (here called $\tilde{u}(u)$), while now v also changed to $\tilde{v}(v)$.

[5] Ignoring the back reaction on the metric.

[6] I have not seen anywhere in the literature a different normalization.

Now, following the same procedure as in section 2.4.1, assume that the renormalized stress tensor is zero when using the coordinates $\tilde{u}$ and $\tilde{v}$. We can calculate the renormalized stress tensor in coordinates u and v by point-splitting regularization and subtracting the stress tensor for $\tilde{u}$ and $\tilde{v}$. This gives

$$\langle T_{uu}\rangle_{\text{ren}} = -\frac{1}{16\pi}\mathcal{F}(\dot{\tilde{u}}; u), \tag{2.21a}$$

$$\langle T_{vv}\rangle_{\text{ren}} = -\frac{1}{16\pi}\mathcal{F}(\dot{\tilde{v}}; v), \tag{2.21b}$$

$$\langle T_{uv}\rangle_{\text{ren}} = \langle T_{vu}\rangle_{\text{ren}} = 0. \tag{2.21c}$$

Dots denote differentiation with respect to the argument.

In order to find the stress tensor in a general spacetime, we must know which coordinates $\tilde{u}$ and $\tilde{v}$ we expect the stress tensor to be zero in. In flat Minkowski space, we assume that an inertial observer will measure a vanishing stress tensor. An accelerating observer, on the other hand, detects a thermal spectrum of particles with a temperature proportional to his proper acceleration. This is known as the Unruh effect. See Unruh [15], or Davies [16] who it would seem was the first to publish.

With this as inspiration, we take $\tilde{u}$ and $\tilde{v}$ to be inertial coordinates in the spacetime. They are given by [14]

$$\tilde{u} - \tilde{u}_0 = \int_{u_0}^{u} \frac{C(u', v_0)}{C(u_0, v_0)} du',$$

$$\tilde{v} - \tilde{v}_0 = \int_{v_0}^{v} \frac{C(u_0, v')}{C(u_0, v_0)} dv'.$$

Which gives[7] from (2.21)

$$\langle T_{uu}\rangle_{\text{ren}} = -\frac{1}{16\pi}\mathcal{F}(C(u, v_0); u), \tag{2.22a}$$

$$\langle T_{vv}\rangle_{\text{ren}} = -\frac{1}{16\pi}\mathcal{F}(C(u_0, v); v), \tag{2.22b}$$

$$\langle T_{uv}\rangle_{\text{ren}} = \langle T_{vu}\rangle_{\text{ren}} = 0. \tag{2.22c}$$

We want the specific case of the space surrounding a planet. The metric is in this case that of Schwarzschild spacetime (cf. (2.6))

$$C(u, v) = 1 - \frac{r_s}{r(u, v)},$$

with $r(u, v)$ implicitly defined through

$$\frac{dv - du}{2} = dr^* = C^{-1} dr,$$

[7] $\mathcal{F}(af; u) = \mathcal{F}(f; u)$ for constant a.

so that

$$\partial_u C(r) = \frac{\partial C}{\partial r}\frac{\partial r}{\partial r^*}\frac{\partial r^*}{\partial u} = -\frac{C}{2}\partial_r C,$$
$$\partial_v C(r) = \frac{\partial C}{\partial r}\frac{\partial r}{\partial r^*}\frac{\partial r^*}{\partial v} = \frac{C}{2}\partial_r C,$$

and in the end, from (2.22)

$$\langle T_{uu}\rangle_{\text{ren}} = \langle T_{vv}\rangle_{\text{ren}} = \frac{1}{16\pi}\left(\frac{1}{4}\frac{r_s^2}{r^4} - \frac{1}{3}\frac{r_s}{r^3}\right),$$
$$\langle T_{uv}\rangle_{\text{ren}} = \langle T_{vu}\rangle_{\text{ren}} = 0.$$

This is not conserved because $\nabla^\mu T_{\mu\nu} \neq 0$, which would violate the conservation of energy and momentum. We therefore impose the conservation of the stress tensor $\nabla^\mu T_{\mu\nu} = 0$, which yields [14]

$$\langle T_{uu}\rangle_{\text{ren}} = \langle T_{vv}\rangle_{\text{ren}} = \frac{1}{16\pi}\left(\frac{1}{4}\frac{r_s^2}{r^4} - \frac{1}{3}\frac{r_s}{r^3}\right), \tag{2.23a}$$
$$\langle T_{uv}\rangle_{\text{ren}} = \langle T_{vu}\rangle_{\text{ren}} = -\frac{1}{48\pi}\left(1 - \frac{r_s}{r}\right)\frac{r_s}{r^3}. \tag{2.23b}$$

The infamous Weyl anomaly of a massless scalar field in two-dimensional curved spacetime is present in the form of the tracelessness-lessness of the stress tensor $\langle T^\mu{}_\mu\rangle_{\text{ren}} \propto \langle T_{uv}\rangle_{\text{ren}} \neq 0$, see Birrell and Davies [8] chapter 6.

This "vacuum polarization" describes a static negative energy density outside of the event horizon, [8] chapter 8. It is usually discussed in the context of black holes, but it should also be valid for horizonless objects in the space described by the Schwarzschild metric. As $r \to \infty$, $\langle T_{\mu\nu}\rangle_{\text{ren}} = 0$ and so there is no Hawking radiation in this case.

The stress tensor from (2.20) is zero when the planet does not collapse. This does not agree with (2.23). These two contributions must be added together to get the complete stress tensor in the case of collapse. This accounts for both the background stress tensor due to the curvature of spacetime known as vacuum polarization and also the radiation emitted due to collapse. Since (2.20) is valid in the far future when the object has essentially collapsed to a black hole, the addition of (2.20) and (2.23) gives the stress tensor of an evaporating black hole. The result is

$$\langle T_{uu}\rangle_{\text{ren}} = \frac{1}{16\pi}\left(\frac{1}{4}\frac{r_s^2}{r^4} - \frac{1}{3}\frac{r_s}{r^3} + \frac{1}{12r_s^2}\right), \tag{2.24a}$$
$$\langle T_{vv}\rangle_{\text{ren}} = \frac{1}{16\pi}\left(\frac{1}{4}\frac{r_s^2}{r^4} - \frac{1}{3}\frac{r_s}{r^3}\right), \tag{2.24b}$$
$$\langle T_{uv}\rangle_{\text{ren}} = \langle T_{vu}\rangle_{\text{ren}} = -\frac{1}{48\pi}\left(1 - \frac{r_s}{r}\right)\frac{r_s}{r^3}. \tag{2.24c}$$

The expression was first calculated by Davies, Fulling, and Unruh [12]. Close to the horizon, it describes ingoing negative energy. The Hawking radiation does not originate at the horizon as in (2.20), it originates in the surrounding space as positive energy going out and an equal amount of negative energy going in [12]. The black hole evaporates, at least qualitatively, according to the ingoing instead of the outgoing Vaidya metric close to the horizon, see section 3.1.1.

The stress tensors in (2.20) and (2.24) are identical in the limit $r \to \infty$. Collapse and evaporation will therefore look identical to an observer far away; "far away" in this context means a distance greater than a few Schwarzschild radii. It is only close to the horizon that the models differ qualitatively in their description of black hole evaporation. This is the topic of chapter 3.

Chapter 3

The Vaidya Metric

Knowing that black holes evaporate, it is of interest to model collapse while accounting for the back reaction on the metric. The simplest such model uses the outgoing Vaidya metric. A collapsing object radiates energy analogous to an accelerating charge in classical electrodynamics. The most interesting quality of this model is that event horizons never form. Kawai *et al.* [3] and also Baccetti *et al.* [4, 5] use the outgoing Vaidya metric to model the formation and evaporation of a black hole[1]. Alternatively, evaporation could be due to the absorption of negative energy. In this case, the ingoing Vaidya metric should be used as the model.

3.1 The Vaidya metric

The Vaidya metric aims to describe the spacetime of a black hole with a changing mass. A first guess for this metric would be to take the Schwarzschild metric (1.3) and make the mass (Schwarzschild radius) time-dependent,

$$ds^2 = -\left(1 - \frac{r_s(t)}{r}\right) dt^2 + \frac{1}{1 - \frac{r_s(t)}{r}} dr^2.$$

The reason this does not work is that it violates relativity. Everyone, whether close to the black hole or far away, will see the mass change simultaneously, even though the information should propagate at the speed of light. The solution is to re-express the line element in terms of null coordinates and make the mass depend on either the outgoing u or ingoing v coordinate (cf. (2.5)).

$$du = dt - \frac{dr}{1 - \frac{r_s}{r}},$$
$$dv = dt + \frac{dr}{1 - \frac{r_s}{r}}.$$

[1] Would-be black hole.

The outgoing and ingoing Vaidya metrics are then given by

$$ds^2 = -\left(1 - \frac{r_s(u)}{r}\right) du^2 - 2dudr, \tag{3.1a}$$

$$ds^2 = -\left(1 - \frac{r_s(v)}{r}\right) dv^2 + 2dvdr. \tag{3.1b}$$

The outgoing Vaidya metric (top) models the mass of the black hole as changing by the emission of null dust. The ingoing Vaidya metric (bottom) models the mass of a black hole as changing by the absorption of null dust.

3.1.1 The outgoing Vaidya metric

The stress tensor for the outgoing Vaidya metric has only one non-vanishing component in null coordinates, T_{uu}. See Poisson [17] chapter 4. It therefore dynamically models black hole evaporation in the case that the stress tensor is given by (2.20), i.e. a black hole which emits radiation.

Taking the outgoing Vaidya metric to model a black hole that evaporates in finite time, it is possible to show that nothing passes the horizon [18]. From (3.1a), an ingoing light ray moves with the speed

$$\frac{dr}{du} = -\frac{1}{2}\left(1 - \frac{r_s(u)}{r}\right).$$

It is easy to check that for all $\frac{dr_s}{du} < 0$ there exists an $r > r_s$ such that $\left|\frac{dr}{du}\right| < \left|\frac{dr_s}{du}\right|$. This means that light cannot fall behind the horizon, and it follows that nothing can. I want to emphasize that it does not matter how slowly the evaporation happens. Unruh [19] makes the point that "pre-Hawking radiation" is such an unfathomably small effect that it could not possibly prevent collapse to the point of horizon avoidance. When using the outgoing Vaidya metric as the model this is wrong.

3.1.2 The ingoing Vaidya metric

Just like the outgoing Vaidya metric the stress tensor for the ingoing Vaidya metric has only one non-vanishing component, in this case T_{vv}. see Poisson [17] chapter 5. The ingoing Vaidya metric does not then model evaporation by the stress tensor in (2.24). Close to the horizon ($r \approx r_s$) it does however take this form. The ingoing Vaidya metric should therefore, in this case, accurately model the back reaction on the metric close to the horizon.

Assuming that the black hole evaporates by the absorption of negative energy null dust, we use the ingoing Vaidya metric. In this case, falling behind the horizon and hitting the singularity is possible. An ingoing light ray will, by definition, travel with constant v. The mass of the black hole is in this case only dependent on v, and so the light ray sees a static black hole.

It is well known that the horizon of a static black hole is transversable, see section 1.2. Thus, the ingoing Vaidya metric does not predict the horizon avoidance of the outgoing Vaidya metric.

Simply using the outgoing Vaidya metric, Kawai *et al.* [3] and Baccetti *et al.* [4, 5] are able to claim that a collapsing shell evaporates before it can cross the horizon. Accounting for the stress tensor being non-zero in curved spacetime, Chen *et al.* [20] unequivocally state that "Black holes do form".

3.2 Collapse and evaporation with the outgoing Vaidya metric

There is another problem with the models discussed in [3, 4, 5]. The claim is made that a collapsing shell of zero thickness will evaporate before crossing the horizon when modelled by the outgoing Vaidya metric. This is not entirely true. Chen *et al.* [20] show that if such a massive shell collapses, the evaporation will cause it to accelerate. So much, in fact, that it will become tachyonic[2] in finite time. This is unphysical, and they conclude that such a shell will have to stop evaporating before the speed of light is reached. This is before the shell has had time to completely evaporate. The model does not prevent the formation of a horizon, and it even predicts that black holes do not radiate after settling.

The results of Kawai *et al.* [3] seem to agree with this. They find that a zero-thickness null shell will emit some radiation in the middle stage of collapse, but stop when the shell approaches the horizon. It is only when they consider a shell of finite thickness that the shell evaporates completely, and the model predicts horizon avoidance. The case of a massive shell of finite thickness has, to my knowledge, yet to be studied.

Baccetti *et al.* [5] also consider a massive shell, however, they did not find that it becomes tachyonic like Chen *et al.* [20] did. In fact they found that it evaporates completely without ever forming a horizon. I have not studied the models sufficiently to comment on why they differ in this very important aspect.

[2] Something that exceeds the speed of light.

Chapter 4

Exotic Compact Objects (ECOs)

Mersini-Houghton [21, 22] posits that the formation of a horizon can be avoided, even if evaporation is caused by the absorption of negative energy, when the ingoing Vaidya metric is appropriate. The claim is made that a collapsing object will, instead of forming a horizon, settle into a compact ball marginally larger than its Schwarzschild radius. These exotic compact objects, ECOs for short, are the black holes in Mersini-Houghton's theory. It is, therefore, interesting to look into whether or not these objects give off Hawking radiation. In this chapter, I derive the relationship between null coordinates u and v in a spacetime with a collapsing shell. This allows for the study of the radiation produced given different collapse trajectories, and for ECOs.

4.1 The function $u(v)$ for a collapsing shell

I will now derive the relationship between null coordinates u and v for a collapsing spherically symmetric shell, following the approach of Birrell and Davies [8] chapter 8. A shell is chosen, so that the interior metric is as simple as possible. It is given by flat Minkowski space. The answers obtained will not differ qualitatively from the more general results obtained from a collapsing ball. On the outside, the metric is given by the Schwarzschild metric (2.6)

$$ds^2 = -C(r)dudv,$$

with $C(r)$ the Schwarzschild factor (1.5)

$$C(r) = 1 - \frac{r_s}{r},$$

and null coordinates defined in (2.5):

$$du = dt - C^{-1}dr,$$
$$dv = dt + C^{-1}dr.$$

On the inside, we have a flat metric

$$ds^2 = -dUdV,$$

with

$$dU = d\tau - dr,$$
$$dV = d\tau + dr.$$

Where τ is the proper time of a stationary observer inside the shell.

Let the radial position of the shell be given by $R(\tau)$, which I refer to as the "collapse trajectory". The two metrics must then be equal at $r = R(\tau)$, thus

$$dUdV = C(R(\tau))dudv. \tag{4.1}$$

Let us determine the relationship between outgoing coordinates u and U and incoming coordinates v and V. We have

$$\frac{du}{dU} = \frac{du}{d\tau}\left(\frac{dU}{d\tau}\right)^{-1} = \frac{\dot{t} - C^{-1}\dot{R}}{1 - \dot{R}}, \tag{4.2a}$$

$$\frac{dV}{dv} = \frac{dV}{d\tau}\left(\frac{dv}{d\tau}\right)^{-1} = \frac{1 + \dot{R}}{\dot{t} + C^{-1}\dot{R}}, \tag{4.2b}$$

where dot denotes the derivative with respect to τ, and $\dot{t}$ is determined with the help of (4.1), from which

$$\dot{t} = C^{-1}\sqrt{C + \dot{R}^2(1 - C)}.$$

Now Birrell and Davies [8] go on to make approximations in order to determine $u(v)$ near the horizon. This is not necessary, and $u(v)$ can be determined exactly by

$$\frac{du}{dv} = \frac{du}{dU}\frac{dU}{dV}\frac{dV}{dv} = \frac{1 + \dot{R}}{1 - \dot{R}}\frac{\sqrt{C + \dot{R}^2(1 - C)} - \dot{R}}{\sqrt{C + \dot{R}^2(1 - C)} + \dot{R}}, \tag{4.3}$$

where $\frac{dU}{dV} = 1$ because they are null-coordinates in a flat spacetime, see (2.2).

For a collapsing shell $\dot{R} \in [-1, 0]$, and it follows that

$$\frac{du}{dv} \in [1, C^{-1}].$$

$u(v)$ is then a bijective, increasing function of v as long as the shell does not cross the horizon ($R(t) > r_s$).

4.2 Simple cases

The differential equation (4.3) is not particularly transparent, and it may be difficult to see what different collapse trajectories will yield in terms of the function $u(v)$ and also Hawking radiation in terms of (2.19). Therefore, I will consider the simple cases of the adiabatic limit, null shells, and a massive shell crossing the horizon. I remind the reader of (2.19) which tells us how much energy is emitted due to a relationship between null coordinates u and v:

$$\langle T_{uu}\rangle_{\text{ren}} \propto \mathcal{F}(\dot{v}(u), u) = \frac{\ddot{v}^2}{\dot{v}^2} - \frac{2}{3}\frac{\dddot{v}}{\dot{v}},$$

where I have substituted the identity $f(u) = v(u)$.

4.2.1 Static ECOs and the adiabatic limit

Consider the adiabatic limit of (4.3), i.e. slow collapse: $\dot{R} = -\varepsilon$, $\varepsilon \ll 1$. Expanding to first order in ε yields

$$\frac{du}{dv} = 1 + 2\left(\frac{1}{\sqrt{1 - r_s/R(\tau)}} - 1\right)\varepsilon + \mathcal{O}(\varepsilon^2),$$

which gives $u \sim v$ as long as the horizon is not crossed; the approximation breaks down for $R = r_s$.

There is no radiation in the adiabatic limit. This agrees with the Bogolubov conclusions from section 2.3.3, that static objects do not emit Hawking radiation.

4.2.2 Null shell

Let us consider the case of a collapsing null shell. That is, take $\dot{R} = -1$ in (4.3). We have

$$\frac{du}{dv} = C^{-1} = \frac{1}{1 - r_s/R(\tau)},$$

and taking $R(\tau) = r_s - \tau = r_s - \frac{1}{2}(v - v_0)$ gives

$$u(v) = v - 2r_s \ln(v_0 - v) + \text{const.}, \tag{4.4}$$

which, as previously seen yields Hawking radiation. See section 2.2.

If we assume that the null shell suddenly stops close to the horizon. We will see Hawking radiation as long as we see the shell falling. Zero radiation is observed after the shell has settled to a static ECO.

4.2.3 Massive shell

Consider now a massive shell that collapses and forms a horizon. Close to the horizon, we have $C \approx 0$. We also take $\dot{R} = -\nu$ constant[1] during the horizon crossing. Expanding (4.3) to lowest order in C yields

$$\frac{du}{dv} = \frac{1}{C}\frac{4\nu^2}{(1+\nu)^2} + \mathcal{O}(1),$$

and taking $R(\tau) = r_s - \nu\tau$ gives

$$\frac{du}{dv} = -\frac{r_s}{\tau}\frac{4\nu}{(1+\nu)^2} + \mathcal{O}(1).$$

From (4.2b) we get

$$\begin{aligned}\frac{dv}{d\tau} = \frac{dv}{dV}\frac{dV}{dU}\frac{dU}{d\tau} &= \frac{(1+\nu)^2}{2\nu} + \mathcal{O}(C) \\ \implies v - v_0 &= \frac{(1+\nu)^2}{2\nu}\tau + \mathcal{O}(C).\end{aligned}$$

Here I have again used $\frac{dV}{dU} = 1$, and also $\frac{dU}{d\tau} = 1 + \nu$.

The final result is

$$\frac{du}{dv} = -\frac{2r_s}{v - v_0} + \mathcal{O}(1),$$

and

$$u(v) = -2r_s \ln(v_0 - v) + \mathcal{O}(v). \tag{4.5}$$

This is identical to (4.4) in the limit $v \to v_0$. Late-time (large u) Hawking radiation is independent of the speed of collapse provided an event horizon does form. This is consistent with the no-hair theorem as the Schwarzschild radius r_s completely determines the renormalized stress tensor. See section 2.4.3.

4.3 Implications

We have seen that ECOs can give off radiation, however they must be collapsing for this to occur. A static ECO will not give off radiation. A theory that predicts horizon avoidance by collapsing matter settling to a static ECO cannot explain black hole evaporation, as the model predicts that no radiation is emitted. See section 5.1 for a possible counterargument.

Having a formula for the exact relationship between null coordinates u and v from (4.3) allows for exact knowledge of how much radiation is emitted during collapse due to the stress tensor (2.19) being wholly determined by

[1] From the equivalence principle.

this. It is the collapse trajectory, not the force of gravity and not the formation of an event horizon, which generates the radiation. Moreover, as seen in (4.5) and (4.4), the collapse trajectory beyond the event horizon does not influence particle production. Thermal Hawking radiation only exists asymptotically, as the function $u(v)$ becomes logarithmic. See section 2.2. As (4.5) shows, this logarithmic relationship is uniquely determined by the Schwarzschild radius r_s and a general feature of all trajectories close enough to the horizon. This is what Barceló *et al.* [23] claim to be the minimal condition for the existence of Hawking radiation. The horizon is not necessary for the emission of radiation, however, it plays an important part in generating thermal Hawking radiation. As the formation of the event horizon is never observed by an external observer, thermal Hawking radiation is also never observed. Radiation emitted by astrophysical black holes will be due to the collapse trajectory prior to horizon crossing. Whether an event horizon forms is not observable, even by the observation of Hawking radiation.

Chapter 5

Advanced Topics and Developments

In this chapter, I briefly discuss some theoretical and also experimental advancements that have been made. Firstly I discuss Schwinger-like particle production for a gravitational field. I then shift the focus over to two-dimensional dilatonic quantum gravity. I conclude the chapter by introducing acoustic black holes for modelling Hawking radiation in the laboratory.

5.1 Gravitaional pair production

A famous example of a non-perturbative result of quantum field theory is the Schwinger effect which predicts pair production in strong electric fields, see Schwinger [24] and Parker and Toms [7] for a textbook treatment. Recently, in 2023, Wondrak *et al.* [25] proposed that a similar pair production effect occurs due to strong tidal forces in gravitational fields. The radiation produced by this mechanism differs from conventional Hawking radiation by its presence even for static objects like planets or static ECOs. I therefore believe this to be a separate effect from Hawking radiation, and not simply a different approach to the calculation.

The basic idea is to find the effective action W by performing a path integral over the field ϕ. Which is given by

$$e^{-iW} = \int \mathcal{D}\phi e^{-iS[\phi]},$$

where the action S is given by the Klein-Gordon action in curved spacetime

$$S[\phi] = \frac{1}{2}\int d^4x\sqrt{-g}\phi\Box\phi = \frac{1}{2}\int d^4x\phi\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi).$$

Performing the path integral gives the more or less well-known result

$$W = \frac{i}{2}\,\mathrm{Tr}\ln\left[\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu)\right],$$

where the path integral has been related to the determinant of the differential operator $\Box$. Using heat-kernel methods, Wondrak *et al.* [25] claim that this leads to black holes radiating with a power

$$P = \frac{9}{40960\pi M^2},$$

about three times higher than the power predicted by Hawking radiation alone (cf. (1.2)). Interestingly, the inclusion of this separate effect does not change the evaporation rate qualitatively. The black hole still evaporates at a rate proportional to M^{-2}.

Because the radiation turns out to be derived from derivatives of the metric tensor $g_{\mu\nu}$, the model predicts radiation also from static objects. This effect will be vanishingly small for objects significantly larger than their Schwarzschild radii like planets or stars, however, it will be about the same for black holes and static ECOs. Theoretically, it would be possible to distinguish black holes from static ECOs by their evaporation rate, as a black hole would then be expected to evaporate more quickly.

Whether this gravitational pair production effect genuinely occurs is still up for debate. Ferreiro *et al.* [26] claim that the effective action is only calculated to first order which should not yield the results claimed by Wondrak *et al.* [25]. Wondrak *et al.* [27] reply to this criticism. I am not familiar enough with the mathematics to comment on the validity of the model. I will however say that it makes intuitive sense that strong tidal forces from a gravitational field will produce particles when it is known that a strong electric field will.

5.2 Dilaton gravity

In this thesis, I have only considered quantum field theory in curved spacetime. Simply put, gravity has been treated as the classical theory of general relativity, while the only quantum aspect has been the matter field ϕ. There is a good reason for this. In order to consider a quantum theory of gravity, one must quantize the Einstein-Hilbert action S_{EH}. In 4 dimensions it is given by

$$S_{\mathrm{EH}} = \frac{1}{8\pi G}\int d^4x\sqrt{-g}R,$$

where G is Newton's constant, g is the determinant of the metric, and R is the Ricci scalar.

This action is non-renormalizable by power counting. g is unitless, R has mass-dimension $[m^2]$, and the measure d^4x has mass-dimension $[m^{-4}]$. For the action to be unitless, the coupling G must have mass-dimension $[m^{-2}]$, which means that an infinite number of counter terms is needed to cancel the divergence of higher-order loops.

In two dimensions the measure is d^2x with mass-dimension $[m^{-2}]$, and so G is unitless. This makes gravity a renormalizable theory in two dimensions. A model of interest is what is known as the CGHS model [28], conformal matter coupled to dilatonic gravity, the action of which is given by

$$S_{\text{CGHS}} = \frac{1}{2\pi}\int d^2x\sqrt{-g}[e^{-2\phi}(R + 4(\nabla\phi)^2) + 4\lambda^2) - \frac{1}{2}(\nabla f)^2].$$

ϕ is here known as the dilaton field, while f is a matter field. λ is the cosmological constant.

This model is of interest because it contains black holes as well as Hawking radiation. The stress tensor takes a form similar to (2.24), i.e. including vacuum polarization, but the Hawking flux is independent of mass. When considering a theory with many matter fields f_i, the would-be black hole evaporates completely before forming a horizon or a singularity [28]. This is surprising given that the metric should be qualitatively equivalent to the ingoing Vaidya metric, see section 3.1.2.

5.3 Black holes in the laboratory

In 1981, Unruh [29] showed that the equation of motion for sound waves in a spherically symmetric, convergent fluid flow[1] is equivalent to that of a massless scalar field in a spacetime with metric given by

$$ds^2 = -(c^2 - v_0^2)d\tau^2 + \frac{cdr^2}{c^2 - v_0^2}. \tag{5.1}$$

Here, c is the speed of sound in the fluid and $v_0(r)$ is the velocity of the fluid and a function of the distance from the centre of the flow r. The "time" τ is defined by

$$d\tau = dt + \frac{v_0 dr}{c^2 - v_0^2}.$$

The metric in (5.1) already looks a lot like the Schwarzschild metric (cf. (1.3)). To make the connection clear, assume that the velocity of the fluid v_0 exceeds the speed of sound c at some position r_s. In the vicinity of $r = r_s$, v_0 is then, to first order in $r - r_s$, approximated by

$$v_0 \approx -c + \alpha(r - r_s),$$

with α some positive constant. The metric (5.1) now takes the form

$$ds^2 \approx -2c\alpha(r - r_s)d\tau^2 + \frac{dr^2}{2\alpha(r - r_s)},$$

[1] Think of water flowing down a drain without rotating.

which is equivalent to the near-horizon approximation of the Schwarzschild metric.

This is the basis for what are known as "acoustic" black holes. Experimental evidence for Hawking radiation was thought to be unobtainable due to the extremely slow evaporation rate, see section 1.1. The study of sound in a fluid is much more experimentally viable. Steinhauer [30] observed, in 2014, Hawking-like radiation from an acoustic black hole in a Bose-Einstein condensate. Ten years later, in March of 2024, Švančara *et al.* [31] published their results after successfully creating an acoustic black hole in a superfluid. The ability to study acoustic black holes in the laboratory allows for hypothesis testing, and maybe even the discovery of other phenomena related to black hole evaporation.

Discussion

The most important work done in this thesis has been to understand Hawking radiation. Hawking's original approach was to calculate the Bogolubov coefficients $\beta_{\omega\omega'}$ in the case of a collapsing object [1]. This depends on the relationship between the null coordinates v on past null infinity $\mathscr{I}^-$ and u on future null infinity $\mathscr{I}^+$. Which, in turn, depends on the exact trajectory of the collapse $R(\tau)$ as seen in section 4.1. Hawking radiation then, is not really a gravitational phenomenon. It comes as a consequence of a disturbance in the relationship between null coordinates like a moving mirror does, and not as a consequence of the strength of gravity. A static object like a planet or a static exotic compact object does not emit Hawking radiation as they do not disturb the relationship between null coordinates. The formation of an event horizon is not necessary for the existence of Hawking radiation, it is sufficient that $u(v)$ takes the logarithmic part of (4.4). This is in exact agreement with Barceló *et al.* [23] who claim that the minimal condition for a thermal Hawking-like flux is $v(u)$ being exponential. $u(v)$ can take a very different form (i.e. (2.14)) while still producing radiation, the difference is that this radiation will not have the same thermal spectrum as that originally discovered by Hawking.

The inspiration for this thesis was mainly the question of accounting for the back reaction of Hawking radiation on the metric. It should not be possible to fall into a black hole that evaporates in finite time, as crossing the horizon takes an infinite amount of time as measured by an external observer. If true, the information loss paradox is completely irrelevant. As was found in chapter 3, this is only accurate when evaporation is modelled by the outgoing Vaidya metric. That means that the black hole itself emits the radiation. An inn-falling observer looking to cross the horizon will then be unsuccessful as he will see the black hole shrink as he encounters more and more radiation. If, on the other hand, evaporation is modelled by the ingoing Vaidya metric, crossing the horizon is possible. This is because this metric models evaporation as the absorption of negative energy. An inn-falling observer may, in this case, be successful in crossing the horizon as he will only have to outrun the incoming negative energy. He is under no obligation to encounter any radiation, as opposed to the case of the outgoing Vaidya metric when he must encounter all the radiation. If both evaporation

and horizon crossing occur, the information loss paradox must be addressed or unitarity discarded, see Hawking [2].

It was found in sections 2.4.1 and 2.4.3 that assuming that the stress tensor is zero outside of a static object leads to collapsing objects emitting radiation and that evaporation is then governed by the outgoing metric. As was shown by Davies *et al.* [12], the stress tensor outside of static gravitating objects is actually non-zero. It takes the form of a small and negative energy. Qualitatively this leads to evaporation by the ingoing Vaidya metric. Accounting for the back reaction on the metric due to Hawking radiation does not prevent the formation of a horizon, and thus it is not a valid solution to the information loss paradox, Chen *et al.* [20].

Outlook

Even though a lot of progress has been made since Hawking first discovered that black holes evaporate, questions still remain.

As discussed in chapter 3, the outgoing Vaidya metric only predicts horizon avoidance in the case of a collapsing null shell of finite thickness. Both Kawai *et al.* [3] and Chen *et al.* [20] conclude that shells of infinitesimal thickness do not evaporate completely before horizon crossing. Baccetti *et al.* [5] disagree and claim that a massive shell of zero thickness evaporates completely before horizon crossing. It would be interesting to understand why they arrive at a fundamentally different result than Chen *et al.* [20]. In the case of a null shell, Kawai *et al.* [3] find that a shell of finite thickness does evaporate without crossing its horizon. It would therefore be interesting to investigate whether a massive shell of finite thickness also heals the inconsistencies pointed out by Chen *et al.* [20].

Wondrak *et al.* [25] propose that gravitational radiation also exists analogous to Schwinger's pair production mechanism [24]. Unlike conventional Hawking radiation, this radiation will also be produced due to static objects like a planet or a static ECO. It will be interesting to follow this research in the coming years.

The dilaton gravity model of Callan *et al.* [28] seems to agree with Davies *et al.* [12] in that the stress tensor outside of a static object is nonzero and negative. As seen in section 2.4.3, this leads to the ingoing Vaidya metric being appropriate for modelling evaporation. They still claim that evaporation prevents horizon formation, and the process is unitary. I would like to understand how this can be the case in a model qualitatively described by the ingoing Vaidya metric.

Although real black holes are predicted to radiate so weakly that we almost certainly will never observe it, all hope is not lost for the experimentalists. So-called acoustic black holes [29] can be studied in the lab. The incredible result of such experiments is that acoustic Hawking radiation has been observed. Recently, Švančara *et al.* [31] have been successful in creating an acoustic black hole in a superfluid. A superfluid flowing without resistance is a much better model for spacetime than a classical fluid.

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